

**HOW TO USE THIS SET**

20 questions, mixed difficulty, 6 case-based. Attempt every question before checking the answer key on the last page.

4 Easy

10 Medium

6 Hard

1.**EASY**

Which stomach region shows shallow pits and long, straight glands packed with parietal and chief cells?

- A. Pylorus
- B. Cardia
- C. Fundus/body
- D. Antrum only
- E. None — all regions look identical

2.**EASY**

Which cell secretes both HCl and intrinsic factor?

- A. Parietal cell
- B. Stem cell
- C. Mucous neck cell
- D. Enteroendocrine cell
- E. Chief cell

3.**EASY**

Which cell secretes pepsinogen and gastric lipase, and is predominant at the base of the gastric gland?

- A. Surface mucous cell
- B. Parietal cell
- C. Chief cell
- D. G cell
- E. Stem cell

4.**EASY**

On H&E staining, parietal cells appear:

- A. Pink (acidophilic), due to packed mitochondria
- B. Black due to melanin
- C. Clear with no staining
- D. Blue/purple (basophilic)
- E. Green due to bile pigment

5.**MEDIUM****CASE**

A biopsy shows very deep, long gastric pits with short, coiled, mucus-dominant glands near the duodenal junction, with scattered hormone-secreting cells.

Which region is this, and which two endocrine cell types are characteristically found here?

- A. Body; enteroendocrine cells secreting secretin
- B. Fundus; parietal and chief cells
- C. Cardia; enterochromaffin cells only
- D. Pylorus; G cells (gastrin) and D cells (somatostatin)
- E. Duodenum; Brunner's gland cells

6.

MEDIUM

CASE

A patient develops pernicious anaemia due to autoimmune destruction of a specific gastric gland cell type, resulting in loss of intrinsic factor and impaired vitamin B12 absorption.

Which cell type has been destroyed, and what else does it normally secrete?

- A. Mucous neck cell — alkaline mucus
- B. Enteroendocrine cell — gastrin
- C. Parietal cell — HCl
- D. Chief cell — pepsinogen
- E. Stem cell — no secretory product

7.

MEDIUM

Fundus and body regions of the stomach are histologically:

- A. Clearly distinguishable by pit:gland ratio
- B. Identical, and cannot be reliably separated on a slide
- C. Distinguished by goblet cell density
- D. Distinguished by the presence of Brunner's glands in the body only
- E. Distinguished by teniae coli in the fundus

8.

MEDIUM

CASE

A basophilic cell at the base of a gastric gland is packed with rough endoplasmic reticulum and contains membrane-limited zymogen granules.

What does this cell secrete?

- A. Neutral surface mucus
- B. HCl and intrinsic factor
- C. Somatostatin
- D. Pepsinogen and gastric lipase
- E. Gastrin

9.

MEDIUM

Following mucosal injury, which cell population proliferates to regenerate the gastric epithelium, and where is it located?

- A. Parietal cells, in the neck
- B. Enteroendocrine cells, at the base
- C. Stem cells, in the neck/isthmus
- D. Surface mucous cells only
- E. Chief cells, at the gland base

10.

MEDIUM

Which gastric region has short glands of roughly equal length to shallow pits, and is dominated by mucous cells with relatively few parietal cells?

- A. Antrum only, per this lecture's terminology
- B. Cardia
- C. Fundus
- D. Pylorus
- E. Body

11.

MEDIUM

Which of the following best describes the epithelium lining the stomach surface and gastric pits across all regions?

- A. Simple columnar, secreting mucus and forming tight junctions as part of the acid barrier
- B. Stratified squamous
- C. Pseudostratified columnar
- D. Transitional epithelium
- E. Simple cuboidal

12.

MEDIUM

Where in the gastric gland are enteroendocrine (APUD) cells typically found, and what do they secrete?

- A. Isthmus; hydrochloric acid
- B. Surface; pepsinogen
- C. Neck; intrinsic factor
- D. Isthmus; mucus only
- E. Base; gastrin, endorphins, and serotonin

13.

MEDIUM

Mucous neck cells are best distinguished from surface mucous cells by which feature?

- A. Mucous neck cells are the primary site of HCl production
- B. Mucous neck cells have the highest mitotic rate in the gland
- C. Mucous neck cells secrete less alkaline mucus and have an irregular shape with basal nucleus and apical secretory granules, unlike the tall columnar surface cells
- D. Mucous neck cells are located exclusively in the pylorus
- E. Mucous neck cells are absent from the fundus/body

14.

MEDIUM

The muscularis externa's middle circular layer thickens in which stomach region to form a functional sphincter?

- A. Fundus
- B. Pylorus, forming the pyloric sphincter
- C. Cardia
- D. Body
- E. Esophagogastric junction

15.

HARD CASE

A gastric biopsy from an achlorhydric patient (no gastric acid production) shows preserved chief cells and mucous cells but a marked reduction in one specific cell population, with normal gland architecture otherwise.

Which cell population is most likely reduced, explaining the achlorhydria?

- A. Parietal cells
- B. Surface mucous cells
- C. Enteroendocrine cells
- D. Stem cells
- E. Chief cells

16.

HARD

A biopsy shows shallow pits, long glands, and a mix of large pink cells with central nuclei alongside smaller basophilic cells with abundant RER, with essentially no D or G cells identified in this field. Which stomach region does this most likely represent?

- A. Esophagus, since it lacks gastric pits entirely
- B. Duodenum, since it has no gastric glands
- C. Pylorus, since G/D cells are typically scarce there
- D. Cardia, since it lacks all endocrine cells
- E. Fundus/body, since G and D cells are a pyloric feature and parietal/chief cells dominate here

17.

HARD CASE

Two gastric biopsies are compared: Biopsy X has a pit:gland ratio favouring short pits and long glands; Biopsy Y has a pit:gland ratio favouring long, deep pits and short coiled glands.

Which biopsy is more likely to show gastrin-secreting G cells, and why?

- A. Neither, since G cells are evenly distributed throughout the stomach
- B. Biopsy X, because G cells concentrate in the acid-secreting zone
- C. Biopsy X, because G cells require long straight glands to develop
- D. Biopsy Y, because its pit:gland pattern (long pits, short coiled glands) is characteristic of the pylorus, where G cells are concentrated
- E. Both equally, since pit:gland ratio does not correlate with endocrine cell distribution

18.

HARD

A drug that selectively inhibits chief cell zymogen granule release, without affecting parietal cells, would be expected to primarily reduce:

- A. Stem cell proliferation in the isthmus
- B. Pepsinogen and gastric lipase secretion, with HCl secretion relatively preserved
- C. Gastrin release from G cells
- D. Gastric acid and intrinsic factor secretion
- E. Mucus secretion from surface cells

19.

HARD

Given that stem cells reside in the neck/isthmus and give rise to all other gastric gland cell types, which observation would be most consistent with impaired stem cell function after chronic mucosal injury?

- A. No effect, since gastric epithelium does not require ongoing renewal
- B. Isolated loss of enteroendocrine cells without effect on other lineages
- C. Delayed or incomplete re-epithelialisation of the gastric mucosa after ulceration, since the regenerative source is compromised
- D. Increased pepsinogen secretion from chief cells
- E. Normal replenishment of surface mucous and parietal cells despite ongoing injury

20.

HARD

CASE

A patient on long-term proton pump inhibitor therapy is noted to have gastrin levels that are chronically elevated on routine bloodwork, despite no gastrinoma being found.

Which physiological mechanism, based on the structure-function relationships in this lecture, best explains the elevated gastrin?

- A. Direct stimulation of G cells by the PPI drug itself
- B. Stem cell exhaustion causing compensatory gastrin release
- C. Loss of mucous neck cells directly raising gastrin
- D. Increased chief cell activity feeding back to raise gastrin
- E. Reduced gastric acid (from parietal cell inhibition) removes the normal negative feedback that acid exerts on G-cell gastrin release

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	C	Fundus/body
2	A	Parietal cell
3	C	Chief cell
4	A	Pink (acidophilic), due to packed mitochondria
5	D	Pylorus; G cells (gastrin) and D cells (somatostatin)
6	C	Parietal cell — HCl
7	B	Identical, and cannot be reliably separated on a slide
8	D	Pepsinogen and gastric lipase
9	C	Stem cells, in the neck/isthmus
10	B	Cardia
11	A	Simple columnar, secreting mucus and forming tight junctions as part of the acid barrier
12	E	Base; gastrin, endorphins, and serotonin
13	C	Mucous neck cells secrete less alkaline mucus and have an irregular shape with basal nucleus and apical secretory granules, unlike the tall columnar surface cells
14	B	Pylorus, forming the pyloric sphincter
15	A	Parietal cells
16	E	Fundus/body, since G and D cells are a pyloric feature and parietal/chief cells dominate here
17	D	Biopsy Y, because its pit:gland pattern (long pits, short coiled glands) is characteristic of the pylorus, where G cells are concentrated
18	B	Pepsinogen and gastric lipase secretion, with HCl secretion relatively preserved
19	C	Delayed or incomplete re-epithelialisation of the gastric mucosa after ulceration, since the regenerative source is compromised
20	E	Reduced gastric acid (from parietal cell inhibition) removes the normal negative feedback that acid exerts on G-cell gastrin release

Histology GIT Block · MEDucation by Mattin Majid · Source: GIT Histology LG 2, Dr Hero K. Mustafa

Recalled and source questions are treated as evidence of what gets asked, not as a source of correct answers — verify against your lecture notes.